

Spatio-angular minimum-variance tomographic controller for multi-object adaptive-optics systems

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Multi-object astronomical adaptive optics (MOAO) is now a mature wide-field observation mode to enlarge the adaptive-optics-corrected field in a few specific locations over tens of arcminutes. The work-scope provided by open-loop tomography and pupil conjugation is amenable to a spatio-angular linear-quadratic-Gaussian (SA-LQG) formulation aiming to provide enhanced correction across the field with improved performance over static reconstruction methods and less stringent computational complexity scaling laws. Starting from our previous work [J. Opt. Soc. Am. A 31, 101 (2014)], we use stochastic time-progression models coupled to approximate sparse measurement operators to outline a suitable SA-LQG formulation capable of delivering near optimal correction. Under the spatio-angular framework the wavefronts are never explicitly estimated in the volume, providing considerable computational savings on 10-m-class telescopes and beyond. We find that for Raven, a 10-m-class MOAO system with two science channels, the SA-LQG improves the limiting magnitude by two stellar magnitudes when both the Strehl ratio and the ensquared energy are used as figures of merit. The sky coverage is therefore improved by a factor of ~ 5 . © 2015 Optical Society of America

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1. INTRODUCTION: MULTI-OBJECT ADAPTIVE-OPTICS SYSTEMS

Multi-object adaptive optics (MOAO) is now a well-established AO concept to enlarge the corrected field to several arcminutes when only a few directions of interest exist in the field and need be corrected for [1–4].

Similarly to multi-conjugate AO (MCAO), it relies on tomography to provide a tridimensional wavefront estimation above the telescope and thereupon provide correction on several scientific relevant directions, reaching out for a multiplexing factor of up to 20 on *extremely large telescope* (ELT)-sized systems [5,6].

In a previous paper [1] we have outlined the static *minimum-mean square error* (MMSE) *spatio-angular* (SA) wavefront estimation (time-dependent wavefront evolution not considered) when a Zernike polynomial basis is used and suggested predictive models to overcome temporal lag errors when aiming

for increased sky coverage [7]. In the SA case estimating the tomographic phase explicitly is circumvented; only the aperture plane is estimated to be later least-squares fitted onto the DM—the *Learn&Apply* approach in [2] is similar albeit in a different vector space.

We now move one step forward: we provide extensions to use a zonal basis set coupled to the optimal minimum residual phase error variance linear-quadratic-Gaussian (LQG) controller tailored to MOAO systems. The LQG controller has been presented in the AO context in several works [4,8–10], and here it is designed and optimized for a MOAO system performing open-loop tomography and single-layer optical conjugation. We further propose a minimal state representation together with first-order near-Markovian predictive models that lead to a computational complexity of the order of the static MMSE reconstructors for a multiplexing capability of 20 science targets.

The use of a spatio-angular framework has several advantages over the formulation that estimates the layered phase explicitly:

- (1) the real-time computation is independent from the number of atmospheric layers;
- (2) the analytical spatio-angular covariance matrices can be computed explicitly, thus avoiding interpolation through the ray-tracing operation to obtain the integrated pupil-plane wavefront profile;
- (3) in principle a hybrid data-driven model-based controller can be designed since a subset of the covariance matrices can be acquired directly from the system (i.e., empirically acquired) instead of relying fully on models; alternatively meta-parameters can be identified from which enhanced models are obtained;
- (4) the sparseness of the LQG compounding matrices can still, albeit to a lesser degree, be exploited for real-time savings as noted in [7,11].

The goal of this paper is therefore two-fold:

- (i) fully outline the zonal LQG minimum-variance tomographic controller design to optimize optical performance, particularly in poor signal-to-noise-ratio (SNR) regimes such that sky coverage can be improved upon by guiding on fainter sources; and
- (ii) provide a suitable, ELT-compliant spatio-angular formulation for the Kalman estimator (and static MMSE) that decouples the wavefront reconstruction from the number of atmospheric layers used to model it.

We present results from both simulation and the optical bench of the Raven system, a MOAO science and technology demonstrator installed on the 8 m Subaru telescope [3].

This paper is organized as follows. Section 2 outlines the LQG formulation, Section 3 discusses other SA implementations, namely the static SA reconstructors obtained from the LQG, Section 4 provides sample numerical results, and Section 5 overviews the Raven optical bench and the results obtained in the lab. Section 6 provides the summary and conclusions.

2. SPATIO-ANGULAR LQG FORMULATION

A. Definitions and Assumptions

We define the pupil-plane wavefront $\psi(\rho, \theta, t)$ indexed by the bidimensional spatial coordinate vector $\rho = [\rho_x, \rho_y]$ in direction $\theta = [\theta_x, \theta_y]$ at time t under the hypothesis that the turbulent atmosphere is a sum of L thin layers located in a discrete number of different altitudes h_l as

$$\psi(\rho, \theta, t) = [\mathbf{H}\boldsymbol{\varphi}](\rho, t) = \sum_{l=1}^L \omega_l \boldsymbol{\varphi}_l(\rho + h_l \theta, t), \quad (1)$$

where $\boldsymbol{\varphi}_l(\rho, t)$ is the l th-layer wavefront, ω_l is the l th-layer relative strength, and $\mathbf{H}$ is a geometric optics propagation operator in the near-field approximation (commonly called ray-tracing [12]) that relates the aperture-plane wavefront (WF) to the layered wavefronts by adding and interpolating $\boldsymbol{\varphi}_l(\rho + h_l \theta, t)$ on the aperture-plane ($h_0 = 0$) computational grid [13].

In the following we assume the frozen-flow hypothesis whereby $\boldsymbol{\varphi}(\rho, t + \tau) = \boldsymbol{\varphi}(\rho - \mathbf{v}\tau, t)$ with $\mathbf{v} = [v_x; v_y]$ the wind

velocity vector. A discrete-time, pointwise representation of the WF is also assumed throughout this paper.

We further assume that averaged phase values over the wavefront sensor (WFS) integration period T_s are taken at face value, i.e.,

$$\boldsymbol{\varphi}_k(\rho) = \frac{1}{T_s} \int_{(k-1)T_s}^{kT_s} \boldsymbol{\varphi}(\rho, t) dt, \quad (2)$$

and that the commands are piecewise constant over an entire frame (either synchronous or not with the WFS sampling frames)

$$\mathbf{u}_k \triangleq \mathbf{u}(t), \quad kT_s \leq t < (k+1)T_s, \quad (3)$$

Ensemble averaging is represented by $\langle \cdot \rangle$ with shorthand covariance matrix notation $\Sigma_{u,v} = \langle uv^T \rangle$. Angles are notated in general θ , and guide-star and science directions use α and β , respectively.

B. LQG Regulator Synthesis

The derivation of the LQG for AO applications has been presented in detail elsewhere for either the infinitely fast DM response or otherwise [9]. In this paper we therefore take a shorter path for the sake of conciseness and refer the reader to the bibliographic references on this particular subject for a more in-depth derivation.

The discrete-time LQG regulator minimizes the cost function

$$J(\mathbf{u}) = \lim_{M \rightarrow \infty} \frac{1}{M} \sum_{k=0}^{M-1} (\mathbf{x}^T \mathbf{Q} \mathbf{x} + \mathbf{u}^T \mathbf{R} \mathbf{u} + 2\mathbf{x}^T \mathbf{S} \mathbf{u})_k, \quad (4)$$

where the weighting triplet $\{\mathbf{Q}, \mathbf{R}, \mathbf{S}\}$ is made apparent and will be specified next by developing a MOAO-specific quadratic criterion subject to the state-space model

$$\begin{cases} \mathbf{x}_{k+1} &= \mathcal{A}\mathbf{x}_k + \mathcal{B}\mathbf{u}_k + \mathcal{V}\nu_k \\ \mathbf{s}_{\alpha,k} &= \mathcal{C}\mathbf{x}_k + \mathcal{D}\mathbf{u}_k + \boldsymbol{\eta}_k \end{cases}, \quad (5)$$

where $\mathbf{x}_k$ is the state vector that contains if not the phase directly a linear combination thereof, $\mathbf{s}_{\alpha,k}$ are the noisy measurements provided by the WFS in the GS directions, and ν_k and $\boldsymbol{\eta}_k$ are spectrally white, Gaussian-distributed state excitation and measurement noise, respectively. In the following we assume that $\Sigma_\nu = \langle \nu \nu^T \rangle$, $\Sigma_\eta = \langle \boldsymbol{\eta} \boldsymbol{\eta}^T \rangle$ are known and that $\Sigma_{\eta,\nu} = \langle \boldsymbol{\eta} \nu^T \rangle = 0$.

C. Minimum Residual Phase Variance After Correction

The objective cost function considered for MOAO is the minimization of the aperture-plane residual phase variance for individual science directions $\beta_i \in \mathbb{R}^{1 \times 2}$ leading to the maximization of the Strehl ratio.

If we denote the term $\epsilon_k \triangleq \psi(\rho, \beta_i, kT_s) - \psi^{\text{cor}}(\rho, \beta_i, kT_s)$ the residual phase profile after DM correction with $\psi(\rho, \beta_i, kT_s) = \mathbf{H}_{\beta_i} \boldsymbol{\varphi}(\rho, kT_s)$ the aperture-plane wavefront in a particular direction in the field and $-\psi^{\text{cor}}(\rho, \beta_i, kT_s)$ the DM-produced correction phase in that direction, the mean-square residual piston-removed wavefront error over the telescope aperture is $\sigma_k^2 = \|\epsilon_k\|_{W, L_2(\Omega)}^2 = \epsilon_k^T W \epsilon_k$, where W is a positive-definite weighting matrix that removes the piston contribution over the telescope aperture [14] and L_2 is the Euclidean norm over the aperture Ω .

The integration, reading, processing, and correction in the AO loop are done in discrete time leading to intrinsic lags that customarily take several milliseconds. We will expand the discussion of time delays for synchronous and asynchronous systems in Section 2.E.

Using the definitions of integrated phase and commands in Section 2.A we can now ascertain the discrete-time minimum residual phase variance (MV) criterion

$$J(\mathbf{u}) = \lim_{M \rightarrow \infty} \frac{1}{M} \sum_{k=0}^{M-1} \sigma_k^2$$

$$= \lim_{M \rightarrow \infty} \frac{1}{M} \sum_{k=0}^{M-1} \|\psi_{k+1}(\rho, \beta) - \mathbf{N}\mathbf{u}_k(\rho, \beta)\|_{W, L_2(\Omega)}^2, \quad (6)$$

which neglects any WF dynamics during the integration. The error that is neglected here has been dubbed *insurmountable error* due to the use of averaged variables instead of continuous ones [9] and is negligible for the most common systems operating at high frame-rates.

Provided the phase vector ψ_{k+1} is known the solution to Eq. (6) is straightforwardly shown to be a least-squares fit onto the DM influence functions [9,15]

$$\mathbf{u}_k(\rho, \beta) = \mathbf{F}\psi_{k+1}(\rho, \beta) = (\mathbf{N}^T \mathbf{W} \mathbf{N})^{-1} \mathbf{N}^T \mathbf{W} \psi_{k+1}(\rho, \beta). \quad (7)$$

The latter would have been likewise found from the degenerate control Riccati equation, whose discussion we avert here but can be found, for instance, in [9,16].

We now deal with the general case in which ψ_{k+1} is not known in Eq. (7) but needs to be estimated from linearly related noisy measurements.

The separation principle [17] immediately provides the solution

$$\mathbf{u}_k(\rho, \beta) = \mathbf{F}\hat{\psi}_{k+1|k}(\rho, \beta) = (\mathbf{N}^T \mathbf{W} \mathbf{N})^{-1} \mathbf{N}^T \mathbf{W} \hat{\psi}_{k+1|k}(\rho, \beta), \quad (8)$$

where $\hat{\psi}_{k+1|k}(\rho, \beta) = \hat{\psi}_{k+1|S_k(\alpha)}(\rho, \beta) = E\{\psi_{k+1}(\rho, \beta) | S_k(\rho, \alpha)\}$ is the conditional expectation of $\psi_{k+1}(\beta)$ in the science directions with respect to the sequence of all measurements from (in general noncoincident) GS directions available up to $t = kT_s$, i.e., $S_k = \{s_0, \dots, s_k\}(\rho, \alpha)$ [18]. This conditional expectation is readily available from a Kalman filter [9,10,17]. It relies on the definition of the first-differences state-space model in Eq. (5) whose construction is shown further below.

1. Near-Markovian Time-Progression Model of the First Order

The use of auto-regressive models has been given wide attention using the Zernike polynomials' expansion basis set [4,19] as well as with Fourier modes for which a complex model encodes perfectly spatial shifts and thus frozen flow [20].

We restrict our attention to the near-Markovian first-order time-evolution model [21], which has delivered best overall performance albeit with increased computational complexity [1,7,22] (spatial dependence omitted),

$$\psi_{k+\Delta}(\alpha) = \mathbf{A}\psi_k(\alpha) + \nu_k(\alpha), \quad (9)$$

where $\Delta = \tau/T_s$ is a delay in units of sample step T_s with the transition matrix minimizes the quadratic criterion [1]

$$\mathbf{A} = \arg \min_{\mathbf{A}} \langle \|\psi_{k+\Delta}(\alpha) - \mathbf{A}'\psi_k(\alpha)\|_{L_2(\Omega)}^2 \rangle. \quad (10)$$

The solution of an Eq. (10) is

$$\mathbf{A} = \langle \psi_{k+\Delta} \psi_k^T \rangle \langle \psi_k \psi_k^T \rangle^{-1}(\alpha). \quad (11)$$

Note that we do not remove piston in this equation although we could following the formulae in [23].

By denoting $\psi_k(\rho, \alpha) = \mathbf{H}_\alpha \phi(\rho, kT_s)$ and substituting in Eq. (11) we get

$$\mathbf{A} = \mathbf{H}_\alpha \langle \phi_{k+\Delta} \phi_k^T \rangle \langle \phi_k \phi_k^T \rangle^{-1} \mathbf{H}_\alpha^\dagger$$

$$= \mathbf{H}_\alpha \langle \phi(\rho - \mathbf{v}\tau) \phi(\rho)^T \rangle \langle \phi(\rho) \phi(\rho)^T \rangle^{-1} \mathbf{H}_\alpha^\dagger, \quad (12)$$

where $\psi_{k+\Delta}(\rho, \alpha) = \mathbf{H}_\alpha \phi(\rho - \mathbf{v}\tau kT_s) = \mathbf{H}_{\alpha'} \phi(\rho, kT_s)$, which translates the spatio-angular nature of the computation we are accomplishing since a time translation is factored in as an altitude-dependent angular shift using the frozen-flow hypothesis. Figure 1 pictorially shows the statements made.

Assuming stationarity, the state excitation noise covariance matrix is found for a first-order time-evolution model from the covariance equality (implicit indices dropped out)

$$\langle \psi \psi^T \rangle = \mathbf{A} \langle \psi \psi^T \rangle \mathbf{A}^T + \langle \nu \nu^T \rangle, \quad (13)$$

since $\langle \psi_{k+1} \psi_{k+1}^T \rangle = \langle \psi_k \psi_k^T \rangle = \langle \psi \psi^T \rangle$. The excitation noise covariance matrix is therefore

$$\langle \nu \nu^T \rangle = \langle \psi \psi^T \rangle - \mathbf{A} \langle \psi \psi^T \rangle \mathbf{A}^T. \quad (14)$$

The model driving noise covariance matrix $\langle \nu \nu^T \rangle$ is a key element of the KF design.

2. Measurement Model

The presentation that now takes place restricts itself to the case of the Shack–Hartmann (SH) WFS, which provides the averaged phase gradient over a regular grid of subapertures in the pupil plane [24].

We use a geometrical linear model that we assume valid for the regime of operation it will be submitted to; the model is thus

$$s_k(\alpha) = \mathbf{G}\psi_k(\alpha) + \eta_k(\alpha), \quad (15)$$

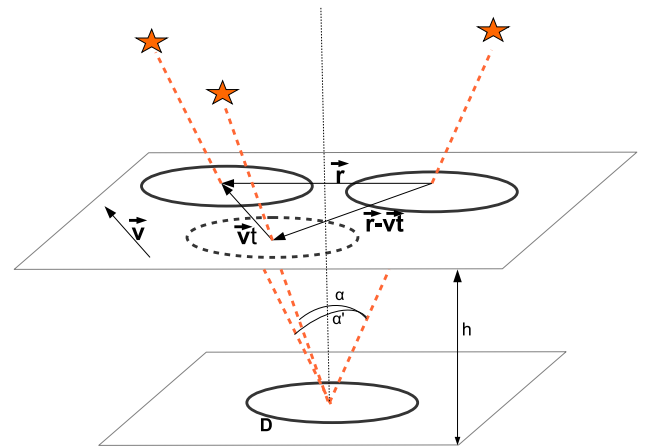


Fig. 1. Spatio-angular computation scheme. Pupil displacements are highly exaggerated for viewing. The spatio-angular correlations are computed with the pupil located at $\rho - \mathbf{v}\tau$, where τ is the time in seconds.

where η_k represents photon and detector shot noise taken to be additive Gaussian noise of known covariance and $\mathbf{G}$ is a matrix whose entries are computed from the discretization of the wavefront gradients in the N_α GS directions. It uses a 3×3 regular stencil to compute the averaged phase gradient with Simpson weights for the fully illuminated subapertures

$$w_x = \frac{1}{d} \begin{bmatrix} -1/4 & 0 & 1/4 \\ -1/2 & 0 & 1/2 \\ -1/4 & 0 & 1/4 \end{bmatrix}, \quad w_y = -w_x^\top. \quad (16)$$

The stencils for partially illuminated subapertures at the outer and inner edges of the telescope pupil are found by expanding the phase on a set of bilinear splines and computing the average gradient over each of the four quadrant patches. The derivation of this result is out of scope but can be found at request in [25].

This sampling and respective estimation of wavefronts at twice the SH-WFS's Nyquist frequency permits the rejection of some aliasing that affects the original measurements as outlined in [19]. An enhanced model with colored noise is shown in [26].

Note that in our implementation we will not make explicit use of the ray-tracing $\mathbf{H}$ in Eq. (15) since the layered phase need not be estimated—no multi-conjugation step ensues. Only the pupil-plane phase is of interest, which opens up to considerable modeling simplifications using a spatio-angular framework.

This is a fundamental simplification with respect to previous tomographic LQG implementations—which in the remainder is called E-LQG to account for the explicit estimation step. We avoid estimating the layered wavefront as is customarily done, focusing on estimating the integrated pupil-plane wavefronts in the GS directions along their lines of sight. The angular anisoplanatic estimate is made on a subsequent step by invoking linearity, stationarity, and Gaussianity as in the static case [1].

D. Minimal State-Space Representation: Compactness in Open-Loop AO

The combination of enhanced spatio-angular first-order predictive models with single-conjugate correction allows for some modeling simplifications that in turn will make for smaller problem size and less stringent real-time and off-line computational requirements.

Under these working assumptions, the LQG model admits—unlike previous implementations [4,22]—a minimal state representation with phase taken at one single time instant. To do so one assumes that the measurements are available at the end of the WFS integration step, regardless of the actual temporal delay in the system [9]: one thus affects the whole delay to the commands instead of the measurements.

On top of this, we can further use a state vector with the pupil-plane phase integrated over directions $\alpha \in [1, \dots, N_\alpha]$ at instant k to provide the *spatio-angular* LQG formulation.

Selecting thus $\mathbf{x}_k \triangleq \boldsymbol{\psi}_k(\alpha)$

$$\boldsymbol{\psi}_k(\alpha) = \begin{pmatrix} \psi(\alpha_1) \\ \vdots \\ \psi(\alpha_{N_\alpha}) \end{pmatrix}_k, \quad (17)$$

where the state is a concatenation of the phase vector in the N_α guide-star directions, one defines the state-space terms for Eq. (9) as

$$\begin{bmatrix} \mathcal{A} & \mathcal{B} & \mathcal{V} \\ \mathcal{C} & \mathcal{D} & 0 \end{bmatrix} = \begin{bmatrix} \mathbf{A} & 0 & \mathbf{I} \\ \mathbf{G} & 0 & 0 \end{bmatrix}. \quad (18)$$

The implementation of the KF involves a real-time state update and prediction equations that in the SA case are

$$\hat{\boldsymbol{\psi}}_{k|k}(\alpha) = \hat{\boldsymbol{\psi}}_{k|k-1}(\alpha) + \mathcal{H}_\infty(\mathbf{s}_k(\alpha) - \mathbf{G}\hat{\boldsymbol{\psi}}_{k|k-1}(\alpha)), \quad (19a)$$

$$\hat{\boldsymbol{\psi}}_{k+1|k}(\alpha) = \mathbf{A}\hat{\boldsymbol{\psi}}_{k|k}(\alpha), \quad (19b)$$

where $\mathcal{H}_\infty$ is the asymptotic Kalman gain computed from the solution of an estimation Riccati equation [9]. The use of the asymptotic value is justified, like elsewhere [27], for one seeks long-exposure performance therefore employing the steady-state gain with no loss of performance. Whenever implicit we drop the notation (α) .

Under the choice of state made earlier, the L_2 criterion in Eq. (6) assumes the equivalent form

$$J(\mathbf{u}) = \lim_{M \rightarrow \infty} \frac{1}{M} \sum_{k=0}^{M-1} \begin{pmatrix} \mathbf{x}_k \\ \mathbf{u}_k \end{pmatrix}^\top \begin{pmatrix} \mathbf{A}^\top \mathbf{A} & -\mathbf{A}^\top \mathbf{N} \\ -\mathbf{N}^\top \mathbf{A} & \mathbf{N}^\top \mathbf{N} \end{pmatrix} \begin{pmatrix} \mathbf{x}_k \\ \mathbf{u}_k \end{pmatrix}, \quad (20)$$

from which $\mathbf{Q} = \mathbf{A}^\top \mathbf{A} \geq \mathbf{0}$, $\mathbf{R} = \mathbf{N}^\top \mathbf{N} > \mathbf{0}$, and $\mathbf{S} = -\mathbf{A}^\top \mathbf{N}$; this construction ensures a stable controller since $\mathbf{A}$ is a positive-definite stability matrix with $|\text{eig}(\mathbf{A})| < 1$ and $\mathbf{Q} - \mathbf{S}\mathbf{R}^{-1}\mathbf{S}^\top \geq \mathbf{0}$ and $\mathbf{R}$ is a full-rank matrix.

E. Time-Delayed Correction: Synchronous and Asynchronous Commands

From phase estimates $\hat{\boldsymbol{\psi}}_{k|k}(\alpha)$ any future disturbance can be computed using the predictive models above for the synchronous and asynchronous correction cases. The latter is dealt with straightforwardly since now the commands are computed for time step $k + \Delta$, for fractional delays of either integer or fractional multiples of the integration time T_s . We further assume that due to the discrete nature of signals and sequential operations, $\tau = T_s + \delta$ leading to $\Delta = \tau/T_s = 1 + \delta/T_s$, i.e., one frame delay plus pure system lag.

Both cases are shown in Fig. 2, providing the temporal sequence of operations.

The controller is applied in real time by computing, at iteration k for either $\delta \geq T_s$ or $\delta \leq T_s$,

$$\hat{\boldsymbol{\psi}}_{k+1|k}(\alpha) = \mathbf{A}\hat{\boldsymbol{\psi}}_{k|k-1}(\alpha) + \mathbf{A}\mathcal{H}_\infty(\mathbf{s}_k(\alpha) - \mathbf{G}\hat{\boldsymbol{\psi}}_{k|k-1}(\alpha)), \quad (21a)$$

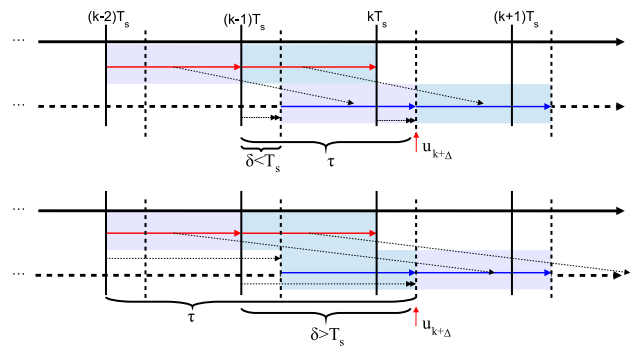


Fig. 2. Temporal diagrams. Top, $\delta < T_s$; bottom, $\delta > T_s$.

$$\mathbf{u}_{k+\delta/T_i}(\beta_i) = \mathbf{F}\Sigma_{\psi\beta, \psi\alpha} \Sigma_{\psi\alpha, \psi\alpha}^{-1} \mathbf{A}_\delta \hat{\psi}_{k+1|k}(\alpha) = \mathbf{T}_\delta \hat{\psi}_{k+1|k}(\alpha), \quad (21b)$$

where $\mathbf{A}_\delta$ estimates the lead wavefront δ seconds ahead in time to overcome any pure lag delay (3 ms for the case of Raven), and multiplication by the anisoplanatic filter $\Sigma_{\psi\beta, \psi\alpha} \Sigma_{\psi\alpha, \psi\alpha}^{-1} = \langle \psi(\beta)\psi(\alpha)^\top \rangle \langle \psi(\alpha)\psi(\alpha)^\top \rangle^{-1}$ provides the pupil-plane wavefront estimate in the science direction of interest. This two-step scheme is equivalent to a single-step prediction on account of the conditional expectation properties, namely

$$E\{\Phi|S=s\} = E\{\Phi|E\{Y|S=s\}\}. \quad (22)$$

The DM projection is made in parallel for all the N_β science directions by multiplying by the generalized inverse of the DM influence matrix in Eq. (7).

3. STATIC AND PREDICTIVE SPATIO-ANGULAR WAVEFRONT ESTIMATION

The state-space framework offers a suitable means to derive static reconstructors against which we will compare the LQG formulation. The presentation that follows is a brief review of the standard minimization criteria in AO that can be found elsewhere [1,14,22,28]. Here we derive it seamlessly from choosing a specific combination of matrices in the LQG controller to set the context for the subsequent simulation results.

A. Static Case

It is a well-established result that the static MMSE solution [1]

$$\hat{\psi}_\beta \triangleq \Sigma_{\psi\beta, s_\alpha} \Sigma_{s_\alpha, s_\alpha}^{-1} s_\alpha \quad (23)$$

$$= \langle \psi_\beta \psi_\alpha^\top \rangle \mathbf{G}^\top (\mathbf{G} \langle \psi_\alpha \psi_\alpha^\top \rangle \mathbf{G}^\top + \langle \eta \eta^\top \rangle)^{-1} s_\alpha \quad (24)$$

can be derived from the LQG solution by taking $\mathcal{A} = 0$ and therefore the conditional expectation $\hat{\psi}_{k|k-1} = 0$ in Eq. (21). In such case the solution of the Riccati equation is $\Sigma_\infty = \Sigma_\varphi$, and the Kalman gain boils down to

$$\mathcal{H}'_\infty = \langle \psi_\beta \psi_\alpha^\top \rangle \mathbf{G}^\top (\mathbf{G} \langle \psi_\alpha \psi_\alpha^\top \rangle \mathbf{G}^\top + \langle \eta \eta^\top \rangle)^{-1}, \quad (25)$$

which provides the same resulting estimate phase vector as in Eq. (25).

B. Predictive Case

To reduce the lag error, spatio-temporal prediction may be built in the reconstructor. In such case one instead computes [29]

$$\hat{\psi}_{k+\Delta}(\beta) \triangleq \langle \psi_{k+\Delta}(\beta) s_k^\top(\alpha) \rangle \Sigma_{s_\alpha, s_\alpha}^{-1}, \quad (26)$$

where $\langle \psi_{k+\Delta}(\beta) s_k^\top(\alpha) \rangle = \langle \psi(\rho - \mathbf{v}\tau, \beta) s^\top(\alpha) \rangle$ is the covariance matrix of the pupil-plane phase between the science and guide-star directions at two time instants separated by a lag τ .

Equation (26) can be split into a spatial estimate followed by a temporal prediction step, being written in the two-step form

$$\hat{\psi}_{k+\Delta}(\beta) = \mathbf{A}_\Delta \hat{\psi}_k(\beta) \quad (27)$$

$$= \mathbf{A}_\Delta \Sigma_{\psi\beta, s_\alpha} \Sigma_{s_\alpha, s_\alpha}^{-1} s_\alpha. \quad (28)$$

Both this two-step approach and the all-at-once approach are equivalent as pointed out before due to conditional expectation

properties. It remains true, however, many steps are made as long as the starting point is a pupil-plane phase estimate.

4. COMPUTATIONAL ASPECTS

A. Off-Line Computation of Spatio-Angular Covariance Matrices

The pointwise covariance matrix computation involves only spatial distances from which the spatio-angular covariance matrices are obtained analytically sampling the covariance function for von-Kármán turbulence

$$C_\psi(\rho) = \left(\frac{L_0}{r_0}\right)^{5/3} \times \frac{\Gamma(11/6)}{2^{5/6} \pi^{8/3}} \left[\frac{24}{5} \Gamma\left(\frac{6}{5}\right)\right] \left(\frac{2\pi\rho}{L_0}\right)^{5/6} K_{5/6}\left(\frac{2\pi\rho}{L_0}\right) \quad (29)$$

with L_0 the outer scale of turbulence, r_0 Fried's parameter, Γ the 'gamma' function, and finally $K_{5/6}$ a modified Bessel function of the third order. Parameter $\rho = |r|$ is the distance between two phase points in the bidimensional plane $\rho = \sqrt{\Delta x^2 + \Delta y^2}$.

For the multiple-layer case, the pupil-plane covariance functions are found from the weighted sum over the layers where the weighting is given by the relative strength of layer l (also known as the fractional r_0).

The spatio-angular covariance matrices are found by sampling and stacking the analytical covariance function in Eq. (29) layer by layer (since layers are considered independent) between points tracing to the WFS subapertures' corners and midpoints whose coordinates in any layer are $[(\rho_{x_{i,n}} + \theta_{x_n} b_l), (\rho_{y_{i,n}} + \theta_{y_n} b_l)]$, and the separation vector between subaperture raster index i on WFS n and subaperture j on WFS m at layer l is

$$\rho_{i,j,l} = (\rho_{i,n} - \rho_{j,m}) + b_l(\theta_n - \theta_m), \quad (30)$$

where the first term on the right-hand side of the equation is the separation between the subapertures in the pupil, and the second term is the global WFS separation vector backprojected in the atmosphere. Figure 3 provides a schematic view of that procedure.

B. Real-Time Computational Burden: Scaling Laws

Regarding computational complexity, the SA-LQG formulation offers a reduced number of real-time operations compared to the E-LQG formulation. Furthermore it provides a less than one order of magnitude increase compared to MMSE (static or predictive) reconstructors for an ELT-sized case with a multiplexing factor of 20.

Take N_β to be the number of DM or likewise science directions, N_{act} to be the number of valid actuators per DM, N_α to be the number of GS directions, $N_s \approx 2N_{\text{act}}$ to be the number of slope measurements per WFS, $N_\varphi = 4N_{\text{act}}$ to be an underestimate of the number of phase points per reconstructed layer, and N_l to be the total number of layers. Table 1 provides the approximate number of real-time multiply-and-accumulate (MAC) operations. If we take the case of Raven with $N_\beta = 2$, $N_{\text{act}} = 97$, $N_\alpha = 3$, and $N_s = 144$, we get an increased computational complexity of

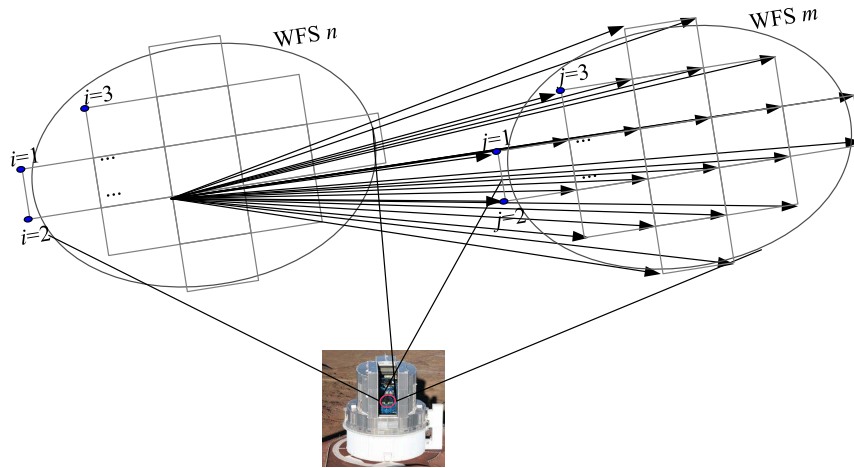


Fig. 3. Spatio-angular computation scheme. Each column in the covariance matrix is obtained by sampling the phase covariance function with distances corresponding to the length of the arrows as shown. To comply with the 3×3 stencil in the gradient operator, the corners and the midpoints are sampled.

Table 1. Approximate Real-Time Computational Complexity

Op./ Algo.	SA-LQG	E-LQG	MMSE
$\hat{\mathbf{C}}_{k k-1}$	$9N_s N_\alpha$	$9N_s N_\alpha +$ $4N_\phi N_l$	
$\mathbf{A}\mathcal{H}_\infty$	$N_\alpha N_\phi N_s N_\alpha$	$N_l N_\phi N_s N_\alpha$	$N_\beta N_{\text{act}} N_\alpha N_s$
$\mathbf{A}\mathbf{x}_{k k}$	$4N_\phi N_\alpha$	$4N_\phi N_l$	
$\mathbf{F}\mathbf{x}_{k+1 k}$	$N_\beta N_{\text{act}} N_\phi$	$N_\beta N_{\text{act}} N_\phi +$ $4N_l N_\phi$	

about a factor 15 for the SA-LQG and 30 for the E-LQG controller with nine layers.

As an example Fig. 4 shows the scaling complexity for ELT-sized systems (e.g., E-ELT MOSAIC [6], TMT's IRMOS [5]) as a function of the number of science targets for $N_{\text{act}} = 3200$

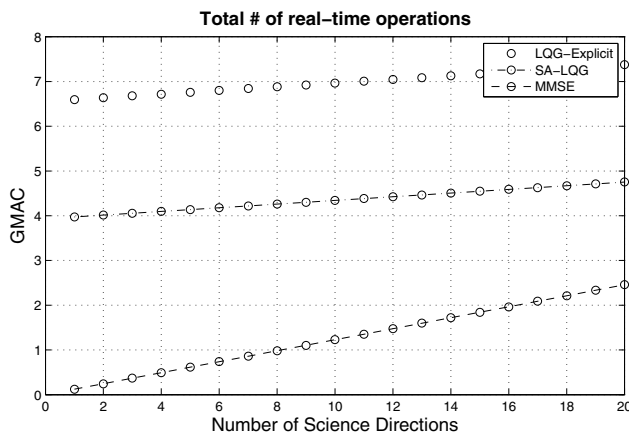


Fig. 4. Number of Giga multiply-and-accumulate (GMAC) real-time operations required by the LQG when compared with the static/predictive MMSE and the explicit LQG. See text for parameters.

(from a 64×64 DM), $N_s = 2N_{\text{act}}$, $N_\phi = 4N_{\text{act}}$, $N_l = 10$, and $N_\alpha = 6$. For a maximum multiplex capacity of 20 science targets SA-LQG and the E-LQG are only a factor of 2 and 3 more demanding than the MMSE, i.e., roughly the same complexity. We have only considered the MMSE implemented as a full VMM since there is growing evidence that this option maps well into GPUs and should therefore be preferred over iterative implementations [30]. For the time being we exclude from this analysis other considerations such as memory bandwidth and pipelinability.

5. SAMPLE NUMERICAL MODELS AND SIMULATIONS

An end-to-end Monte Carlo wave-optics simulation of the Raven tomographic system in a zonal basis has been set up using the object-oriented OOMAO simulator [31] to examine the performance of the SA-LQG algorithm in relation to simpler static and predictive reconstructors.

A. Model Parameters and Validation

The figure-of-merit we use to establish the quality of the wavefront correction is the ensquared energy (EE); this is computed as the ratio of the amount of light falling on a science camera within a given angle projected on-sky to the total amount of light reaching the detector. Strehl ratios are also computed for completeness.

Table 2 contains the main simulation parameters, selected to reflect the physical properties available in the calibration unit of our Raven system test-bench [32]. These include a three-layer atmosphere located at ground, 5.5 and 11 km altitudes, a turbulence coherence length r_0 of 19 cm, and an atmospheric outer scale L_0 of 40 m.

Model behavior was verified by collecting a simulated long-exposure science image using no AO correction mode. The theoretical estimate of the full width at half maximum (FWHM) of the long-exposure PSF for the baseline

Table 2. Raven Baseline Configuration Parameters

Telescope	
D	8 m
Atmosphere	
r_0	19 cm
L_0	40 m
Zenith angle	0 deg
Fractional r_0	[0.596; 0.224; 0.180]
Altitudes	[0, 5.5, 11] km
Wind speeds	[5.68; 6; 17] m/s
Wind direction	[90; 180; 180] deg
Wavefront Sensor	
N_{NGS}	3
Order	10×10
θ_{pix}	0.4 arcsec
RON	0.2 e ⁻
NGS radii	45 arcsec
N_{pix}	12
f_{sample}	30–200 Hz
λ_{WFS}	0.64 μm
Centroiding algorithm	Thresholded center-of-gravity
DM	
N_{DM}	2
Order	11×11
Stroke	infinite
Influence	cubic
AO loop	
Pure delay	$\tau_{\text{lag}} = 3 \text{ ms}$
Evaluation Wavelength	
λ_{evl}	1.65 μm

atmosphere in the J band (1.2 μm) can be computed for a finite outer scale using the expression developed in [33] for the ratio, $L_0/r_0 > 20$. With $r_0 = 0.156 \text{ cm}$ at 0.5 μm and $L_0 = 30 \text{ m}$ (chosen from median conditions on Mauna Kea), this computation yields an expected FWHM of 0.3891 arcsec in the J band. The FWHM of a simulated long-exposure science image was computed to be 0.3865 arcsec, confirming good agreement between theory and simulation. Note that this test was carried out during initial validation of the simulation tools before parameters such as r_0 were finalized.

B. Results

Previous work [1] examined the trade-off between lower WFS frame-rates, which reduce spatial errors due to noise

propagation since the measurement's SNR is higher, and increased temporal lag error. Correction quality was shown to increase with WFS integration time to a point and then begin to decrease as the lag error exceeded the SNR gain.

Table 3 summarizes the peak performance over the frame-rates for varying GS magnitudes.

These tests show that, as in the modal case, better peak performance can be achieved using temporal prediction, and this peak occurs at lower frame-rates than the static algorithm for each magnitude. Providing reduced noise propagation, the SA-LQG algorithm shows that the best performance occurs at approximately the same WFS frame-rate as in the predictive case, but that the overall performance is significantly improved, indicating a reduction in spatial error. The data are also plotted in Fig. 5; here it can be seen that by using temporal prediction, equivalent performance can be achieved with higher-magnitude GSs when using a lower frame-rate. The LQG can deliver performance equivalent to that of the predictive algorithm using GSs that are one magnitude dimmer and equivalent to the static algorithm using GSs that are greater than two magnitudes dimmer.

C. SA-LQG versus E-LQG

In Appendix A it is shown that the SA-LQG can be deduced from the E-LQG (under certain conditions) in a stationary regime.

The geometrical nature of the problem, however, is such that we can retrieve the pupil-plane wavefront from its layered counterpart through ray-tracing but not the opposite. This is patent in the noninvertibility of $\mathbf{H}$, which imposes a spatial truncation on the wavefronts over the footprints in the GS direction otherwise represented on larger meta-pupils.

Simulations have shown that

$$\langle \|\psi_k(\alpha) - \hat{\psi}_{k|k}(\alpha)\|^2 \rangle \geq \langle \|\psi_k(\alpha) - \mathbf{H}_\alpha \hat{\phi}_{k|k}\|^2 \rangle; \quad (31)$$

i.e., the averaged residual phase variance in the GS directions obtained with the SA-LQG is higher than that obtained with the E-LQG. This difference then translates to the science when using the anisoplanatic filter as follows:

$$\langle \|\psi_k(\beta) - \Sigma_{\rho_\beta, \rho_\alpha} \Sigma_{\rho_\alpha, \rho_\alpha}^{-1} \hat{\psi}_{k|k}(\alpha)\|^2 \rangle \geq \langle \|\psi_k(\beta) - \mathbf{H}_\beta \hat{\phi}_{k|k}\|^2 \rangle. \quad (32)$$

The anisoplanatic filter *per se* adds no further error to the estimation since

Table 3. Raven End-to-End Simulation Results^a

GS Mags	Static SA				SA Prediction				SA LQG			
	EE	Lag	Strehl	Lag	EE	Lag	Strehl	Lag	EE	Lag	Strehl	Lag
13.5	47.99	6	28.55	13	49.88	23	31.70	15	52.27	20	33.54	15
14	47.48	10	28.05	14	50.01	25	32.32	22	52.66	22	34.60	22
14.5	46.79	12	27.98	15	49.65	24	31.72	24	52.48	24	34.02	23
15	45.73	13	25.64	15	49.07	25	30.66	25	52.06	25	33.12	25
15.5	44.36	15	23.41	18	48.41	26	29.11	25	51.40	25	31.53	25
16	42.57	16	21.52	22	47.31	30	26.50	25	50.27	26	29.25	27
16.5	40.43	20	18.90	25	46.73	41	24.33	40	48.98	35	26.72	35
17	38.28	25	15.57	25	45.20	46	23.29	47	47.48	45	24.95	45

^aThe optimal performance (percent ensquared energy) for each GS magnitude is shown for the zonal static SA, and compared with standalone SA prediction and the full SA LQG algorithm.

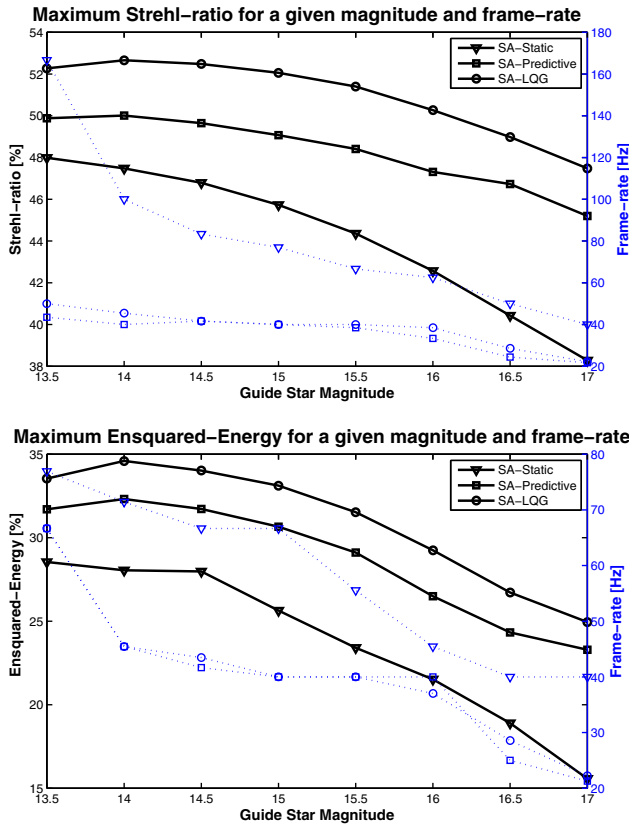


Fig. 5. Static reconstructor versus standalone prediction and LQG: Strehl ratios (top) and ensquared energy (bottom) as a function of NGS magnitude. Left axis shows the best achieved performance with a static reconstructor compared with the predictive and LQG reconstructors; right axis shows the OL-WFS sample rate at which that performance is obtained.

$$\langle \|\psi_k(\beta) - \sum_{\rho_{\alpha}\rho_{\alpha}} \Sigma_{\rho_{\alpha}\rho_{\alpha}}^{-1} \mathbf{H}_{\alpha} \hat{\phi}_{k|k} \|^2 \rangle = \langle \|\psi_k(\beta) - \mathbf{H}_{\beta} \hat{\phi}_{k|k} \|^2 \rangle. \quad (33)$$

A modal decomposition of the errors using Zernike polynomials showed that the low-order modes such as tilts, focus, and astigmatism are more affected. This is also in agreement with the small impact observed in EE, which would suffer more from poorer correction of high-order modes.

For the Raven case, the differences observed in the science direction are of the percent level.

6. RAVEN LABORATORY TEST RESULTS

Raven's opto-mechanical design and system capabilities are summarized in [3,32,34]; it is equipped with a calibration unit (CU), which also serves as a telescope simulator and turbulence generator. A broad spectrum source illuminates an array of pinholes, any three of which can be picked off to create an NGS asterism, and a series of filters that allow incremental changes in the magnitudes of all asterism stars at once. A cross section of measurements across NGS WFS frame-rates was taken for three different filters to show the improvement in performance using prediction and SA-LQG control over the static reconstructor.

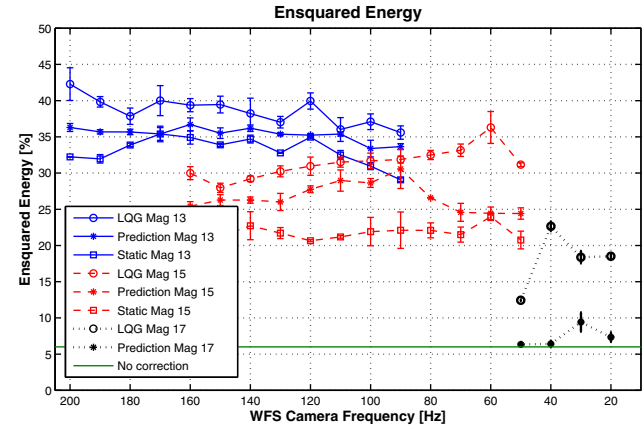


Fig. 6. Measured ensquared energy in a 140 millircsec slit of science images taken while running the NGS WFSs at varying frame-rates and executing static, predictive, and LQG algorithms at three different magnitude settings. The static reconstructor results are not shown for the faintest stars as the performance is not better than with no AO correction.

The laboratory results also validate our simulation results, which indicate that, using the SA-LQG reconstructor, the system can produce performance equivalent to that obtained using the static reconstructor when GSs are two magnitudes fainter. A plot of the results of these measurements is shown in Fig. 6. The diameter of the asterism used is approximately 2 arcmin. The CU lamp does not illuminate all pinholes with uniform intensity; therefore the data points labeled Mag 13 were generated using GSs with magnitudes [12.6, 13.25, 13.25], those labeled Mag 15 are actually [14.3, 15, 15], and those labeled Mag 17 are [16.5, 17.2, 17.2].

The performance achieved in lab tests is lower than that predicted by simulations in all reconstructor cases. However, the trend of increasing peak performance with each increase in reconstructor complexity is reflected in both the simulation data and the measured data. The overall decrease in performance can be attributed to multiple sources; these include imperfect calibration, which has a significant impact on OL systems, underestimation of noise sources in the simulation compared to reality, effects of the rotated WFSs, DM fitting, OL go-to errors, and noncommon path aberrations (NCPAs) between the OL and science paths, as well as between the CL-WFSs and the science camera. A higher amount of total system lag than anticipated may also be reducing the performance of all three types of reconstructors. However, the main simulation findings in improved performance could be reproduced with the test-bench. We are currently investigating ways to enhance the calibration and fine knowledge of system parameters in order to optimize overall system performance and gain understanding of unmodeled system features.

The science images in Fig. 7 represent the best performance achieved for each reconstruction method for a fixed magnitude; they clearly show that the EE is increased and the spot image becomes smaller for both predictive and LQG algorithms over the static reconstructor.

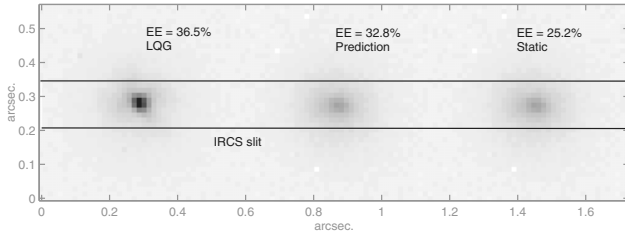


Fig. 7. Comparison of science camera images for SA-LQG, predictive, and static tomographic reconstruction algorithms. Peak frequency PSF shown for each reconstructor using magnitude 15 NGSS (see Fig. 6).

7. SUMMARY

We have outlined the dynamical Strehl-optimal LQG controller providing minimum residual phase variance tailored to MOAO systems. The latter does not require explicit knowledge of the tridimensional wavefront profile allowing for a simplified forward model. Our implementation uses anisoplanatic filters to estimate pupil-plane wavefront profiles from any other directions based on knowledge of the turbulent C_n^2 profile. One such approach can also be useful for laser tomography AO.

We have included temporal prediction by assuming frozen flow, in which case time evolution can be converted to spatial translations. We have proposed a minimal space-state representation with a single temporal instance providing reduced computational complexity.

By allowing the controller to run slower while performing temporal prediction, we can increase the SNR in the wavefront measurements by collecting more light in each SH subaperture. We have shown that we could gain two magnitudes for the same performance obtained with a MMSE static reconstructor, leading to a sky-coverage improvement of roughly a factor of 5.

APPENDIX A: RELATIONSHIPS BETWEEN SA-LQG AND EXPLICIT LQG

We set forth to show that the SA-LQG formulation provided in this paper shares important features with the more standard “explicit” E-LQG formulation, which involves the estimation of phase in the layers. Let the E-LQG state update equation

$$\hat{\phi}_{k+1|k} = \mathbf{A}_E \hat{\phi}_{k|k-1} + \mathcal{L}_\infty^E (s_k - \mathbf{G} \mathbf{H} \hat{\phi}_{k|k-1}), \quad (\text{A1})$$

where upper/subscript ‘E’ indicates matrices for the E-LQG case.

For the near-Markovian order 1 time-progression model assumed earlier, the driving noise statistics are such that $\Sigma_n^E = \Sigma_\varphi - \mathbf{A}_E \Sigma_\varphi \mathbf{A}_E^T$. Multiplying out by the ray-tracing operator to the left and its transpose to the right equates to $\Sigma_n = \Sigma_\varphi - \mathbf{A} \Sigma_\varphi \mathbf{A}^T$ by plausibly taking

$$\mathbf{A} = \mathbf{H} \mathbf{A}_E \mathbf{H}^\dagger \quad (\text{A2})$$

and assuming $\mathbf{H}^\dagger$ to be computable and furthermore that $\mathbf{H} \mathbf{H}^\dagger = \mathbf{I}$. This assumption, however, may not hold since $\mathbf{H}$ is nonsquared and its structure gives rise to a rank deficient matrix. We thus expect (and have evidence for) some

differences using the SA-LQG with respect to the E-LQG formulation. We now add the following:

$$\mathbf{H} \mathbf{A}_E = \mathbf{H} (\mathbf{H}^\dagger \mathbf{A} \mathbf{H}) \quad (\text{A3})$$

$$= \mathbf{A} \mathbf{H} \quad (\text{A4})$$

and

$$\mathbf{H} \mathcal{L}_\infty = \mathbf{H} \mathbf{A}_E \mathcal{M}_\infty^E \quad (\text{A5})$$

$$= \mathbf{A} \mathbf{H} \mathcal{M}_\infty^E \quad (\text{A6})$$

$$= \mathbf{A} \mathcal{M}_\infty. \quad (\text{A7})$$

Multiplying the state update equation on both sides by $\mathbf{H}$ to the left equates to

$$\mathbf{H} \hat{\phi}_{k+1|k} = \mathbf{H} \mathbf{A}_E \hat{\phi}_{k|k-1} + \mathbf{H} \mathcal{L}_\infty^E (s_k - \mathbf{G} \mathbf{H} \hat{\phi}_{k|k-1}) \quad (\text{A8})$$

$$= \mathbf{H} (\mathbf{A}_E - \mathcal{L}_\infty^E \mathbf{G} \mathbf{H}) \hat{\phi}_{k|k-1} + \mathbf{H} \mathcal{L}_\infty^E s_k. \quad (\text{A9})$$

With the above identities one gets $\mathbf{H} \mathcal{L}_\infty^E \mathbf{G} \mathbf{H} \hat{\phi}_{k|k-1} = \mathbf{A} \mathcal{M}_\infty \mathbf{G} \hat{\psi}_{k|k-1}$, yielding

$$\hat{\psi}_{k+1|k} = \mathbf{A} (\mathbf{I} - \mathcal{M}_\infty \mathbf{G}) \hat{\psi}_{k|k-1} + \mathcal{M}_\infty s_k, \quad (\text{A10})$$

which is the SA formulation. We are left with the derivation of the equivalence of Kalman gains and the respective Riccati solutions, both computed off-line. The Riccati equation writes

$$\begin{aligned} \Sigma_\infty^E &= \mathbf{A}_E \Sigma_\infty^E \mathbf{A}_E^T + \Sigma_n^E \\ &\quad - \mathbf{A}_E \Sigma_\infty^E \mathbf{H}^\dagger \mathbf{G}^\dagger (\mathbf{G} \Sigma_\infty^E \mathbf{G}^\dagger + \Sigma_\eta)^{-1} \mathbf{G} \mathbf{H} \Sigma_\infty^E \mathbf{A}_E^T. \end{aligned} \quad (\text{A11})$$

Multiplying by $\mathbf{H}$ on the left and $\mathbf{H}^\dagger$ on the right, one immediately gets

$$\Sigma_\infty = \mathbf{A} \Sigma_\infty \mathbf{A}^T + \Sigma_n \quad (\text{A12})$$

$$- \mathbf{A} \Sigma_\infty \mathbf{G}^\dagger (\mathbf{G} \Sigma_\infty \mathbf{G}^\dagger + \Sigma_\eta)^{-1} \mathbf{G} \Sigma_\infty \mathbf{A}^T, \quad (\text{A13})$$

where we take $\Sigma_\infty = \mathbf{H} \Sigma_\infty^E \mathbf{H}^\dagger$ with the state noise covariance matrix in the SA case obtained from the E-LQG using the definitions above. In practice this assumption does not hold in part due to the finite number of iterations used to compute the asymptotic Riccati estimation error covariance matrix Σ_∞ and its E-LQG counterpart, and in part since the geometry of the problem imposes a spatial truncation on baselines over which covariances are computed unfavorably to the SA-LQG. The Kalman gain can then be obtained as

$$\mathbf{H} \mathcal{M}_\infty^E = \mathbf{H} \Sigma_\infty^E \mathbf{H}^\dagger \mathbf{G}^\dagger (\mathbf{G} \Sigma_\infty^E \mathbf{H}^\dagger \mathbf{G}^\dagger + \Sigma_\eta)^{-1}, \quad (\text{A14})$$

$$\mathcal{M}_\infty = \Sigma_\infty \mathbf{G}^\dagger (\mathbf{G} \Sigma_\infty \mathbf{G}^\dagger + \Sigma_\eta)^{-1}, \quad (\text{A15})$$

which is the straight SA formulation. Further, the results shown in Section 5.C, by using the Kalman gain for the SA-LQG from Eq. (A14), do not change the performance reported.

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